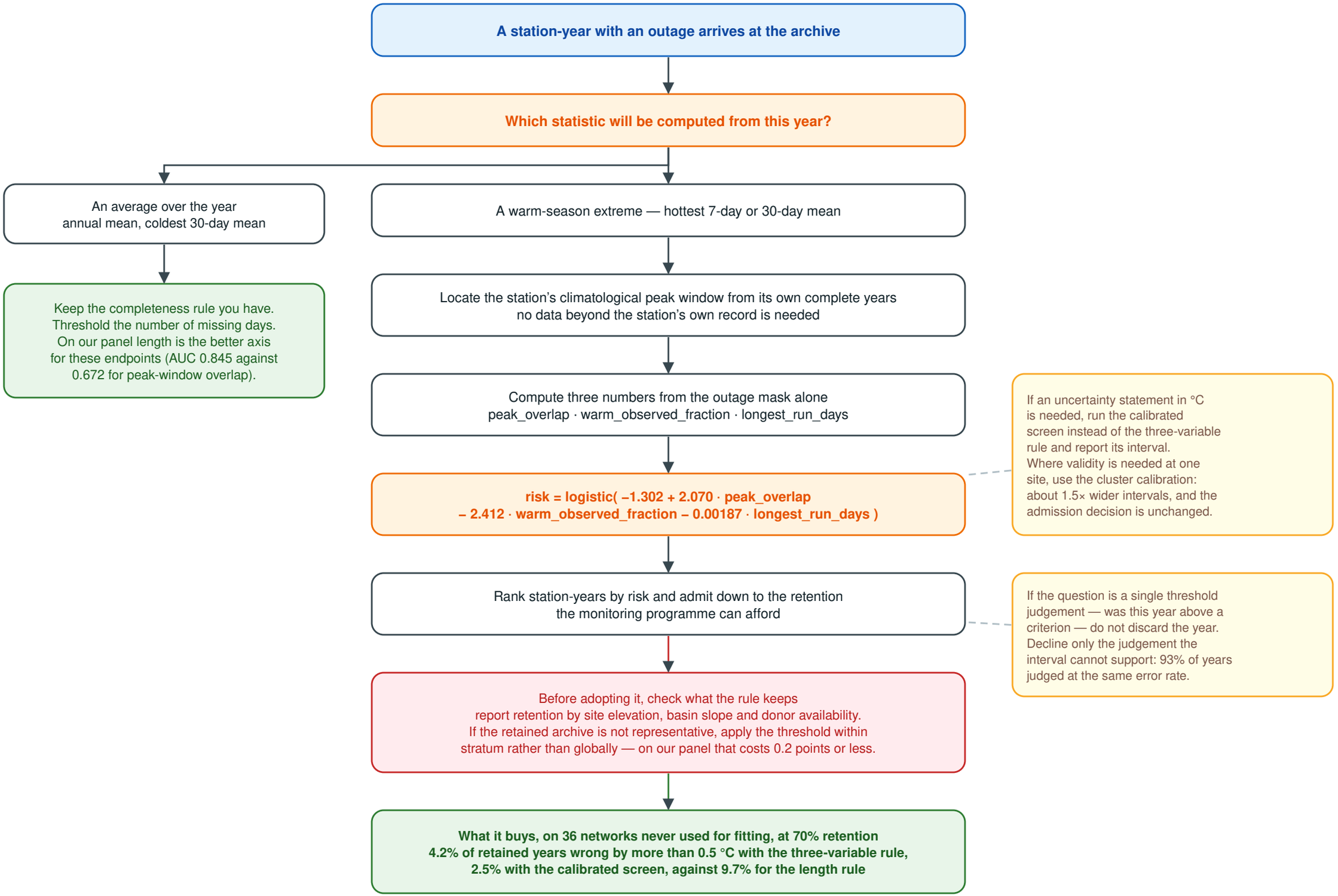
A station-year with an outage arrives at the archive
Which statistic will be computed from this year?
An average over the year
annual mean, coldest 30-day mean
Keep the completeness rule you have.
Threshold the number of missing days.
On our panel length is the better axis
for these endpoints (AUC 0.845 against
0.672 for peak-window overlap).
A warm-season extreme — hottest 7-day or 30-day mean
Locate the station's climatological peak window from its own complete years
no data beyond the station's own record is needed
Compute three numbers from the outage mask alone
peak_overlap · warm_observed_fraction · longest_run_days
risk = logistic(–1.302 + 2.070 · peak_overlap
– 2.412 · warm_observed_fraction – 0.00187 · longest_run_days)
If an uncertainty statement in °C
is needed, run the calibrated
screen instead of the three-variable
rule and report its interval.
Where validity is needed at one
site, use the cluster calibration:
about 1.5× wider intervals, and the
admission decision is unchanged.
Rank station-years by risk and admit down to the retention
the monitoring programme can afford
If the question is a single threshold
judgement — was this year above a
criterion — do not discard the year.
Decline only the judgement the
interval cannot support: 93% of years
judged at the same error rate.
Before adopting it, check what the rule keeps
report retention by site elevation, basin slope and donor availability.
If the retained archive is not representative, apply the threshold within
stratum rather than globally — on our panel that costs 0.2 points or less.
What it buys, on 36 networks never used for fitting, at 70% retention
4.2% of retained years wrong by more than 0.5 °C with the three-variable rule,
2.5% with the calibrated screen, against 9.7% for the length rule